

THE MONOIDAL CENTRE FOR GROUP-GRADED CATEGORIES

BRANKO NIKOLIĆ AND ROSS STREET

ABSTRACT. Let G be a group and k be a commutative ring. Our aim is to ameliorate the G -graded categorical structures considered by Turaev and Virelizier by fitting them into the monoidal bicategory context. We explain how these structures are monoidales in the monoidal centre of the monoidal bicategory of k -linear categories on which G acts. This provides a useful example of a higher version of Davydov's full centre of an algebra.

INTRODUCTION

As well as being a mathematical discipline with its own structures, calculations and significant theorems, category theory provides a framework for understanding analogies between concepts in different disciplines as examples of the same concept interpreted in different categories. Bicategories do the same for categorical concepts, and so on, iterating the microcosm principle (so named by Baez and Dolan; see footnote 2 on page 100 of [8]). An example of this unification process is provided in Section 4.8 of Street [26] where the full centre of an algebra, as defined and constructed by Davydov [6], is shown to be a monoidal centre in the sense of Street [24] in the monoidal bicategory of pointed categories.

We show here how the structures defined by Turaev and Virelizier [27] for G -graded categories, where G is a group, are precisely the usual monoidal structures taken in an appropriate monoidal bicategory. To some extent, we replace the monoidal category of k -modules by the monoidal bicategory Mod^{op} whose morphisms are two-sided modules on which categories act.

There are several benefits gained by working at the bicategorical level. While k -modules only have monoidal duals if they are finitely generated and projective, all objects of Mod^{op} have duals. While k -modules only have finite direct sums, all small direct sums exist in Mod^{op} . While the monoidal centre of the category of all k -linear representations of a group G is not tortile (also called “ribbon”), the monoidal centre of the bicategory of Mod^{op} -representations of G is tortile. Moreover, a higher version of the Davydov full centre is involved in the key example mentioned in [27].

At this stage we are not saying anything about the main theorem of [27] which is a G -graded version of a theorem of Müger [18].

Date: July 26, 2020.

The second gratefully acknowledges the support of Australian Research Council Discovery Grant DP190102432.

1. BASICS ON BICATEGORIES

Bicategories as defined by Bénabou [2] are “monoidal categories with several objects” in the sense that additive categories are “rings with several objects”. For bicategories $\mathcal{A}$ and $\mathcal{B}$, we write $\text{Ps}(\mathcal{A}, \mathcal{B})$ for the bicategory of pseudofunctors, pseudonatural transformations, and modifications (in the terminology of Kelly-Street [15]). We will use monoidal bicategory terminology from Day-Street [8] except that we now use “monoidale” in preference to “pseudomonoid”.

A good example of an autonomous symmetric monoidal bicategory is Mod . The objects are categories. The homcategories are defined by

$$\text{Mod}(\mathcal{A}, \mathcal{B}) = [\mathcal{B}^{\text{op}} \times \mathcal{A}, \text{Set}] ;$$

objects of these homs are called *modules*. Composition

$$\text{Mod}(\mathcal{B}, \mathcal{C}) \times \text{Mod}(\mathcal{A}, \mathcal{B}) \xrightarrow{\circ} \text{Mod}(\mathcal{A}, \mathcal{C})$$

is defined by

$$(N \circ M)(C, A) = \int^B M(B, A) \times N(C, B) .$$

The identity module $\mathcal{A} \rightarrow \mathcal{A}$ is the hom functor of $\mathcal{A}$. Tensor product is finite cartesian product of categories; it is not the product in Mod . Coproduct in Mod is coproduct of categories; it is also product in Mod . We identify each functor $F : \mathcal{A} \rightarrow \mathcal{B}$ with the module $F_* : \mathcal{A} \rightarrow \mathcal{B}$ defined by $F_*(B, A) = \mathcal{B}(B, FA)$; this gives a faithful locally-full pseudofunctor $(-)_* : \text{Cat} \rightarrow \text{Mod}$ which is the identity on objects, strong monoidal, and coproduct preserving. The dual of $\mathcal{A}$ in Mod is the opposite category $\mathcal{A}^{\text{op}}$.

Recall (for example from [8]) that monoidales in Mod^{op} are promonoidal categories $\mathcal{A}$ in the sense of Day [7]. The tensor product is a module $P : \mathcal{A} \rightarrow \mathcal{A} \times \mathcal{A}$ and the unit is a module $J : \mathcal{A} \rightarrow \mathbf{1}$. The Day convolution monoidal structure on the functor category $[\mathcal{A}, \text{Set}]$ has unit $J : \mathcal{A} \rightarrow \text{Set}$ and tensor product $F \star G$ defined by

$$(F \star G)C = \int^{A, B} P(A, B; C) \times FA \times GB .$$

Suppose $\mathcal{V}$ is a complete cocomplete closed symmetric monoidal category. We have a symmetric (weak) monoidal pseudofunctor

$$[-, \mathcal{V}] : \text{Mod}^{\text{op}} \longrightarrow \mathcal{V}\text{-CAT} \quad (1.1)$$

taking each category $\mathcal{A}$ to the $\mathcal{V}$ -enriched functor category $[\mathcal{A}, \mathcal{V}]$. Therefore, each promonoidal category is taken to a convolution monoidal $\mathcal{V}$ -category. In the case $\mathcal{V} = \text{Set}$, (1.1) is the contravariant representable pseudofunctor $\text{Mod}(-, \mathbf{1})$.

Let G be a group regarded as a one-object category. The object will be denoted by o_G . The morphisms of this category will also be called *elements* of G ; so $a \in G$ will mean, as usual, that a is an element. We also use the notation ${}^c a = cac^{-1}$ and $a^c = c^{-1}ac$.

Like every category, G has a promonoidal structure for which the convolution structure on $[G, \text{Set}]$ is pointwise cartesian product. In this case, the unit J is constant at the one-point set and the module $P : G \rightarrow G \times G$ is defined by

$$P(o_G, o_G; o_G) = \{(x, y) : x, y \in G\} , \quad P(a, b; c)(x, y) = (cxa, cyb) . \quad (1.2)$$

As we expound the theory, we will carry along an example based on Example 3.5 of [27] which, in turn, was based on an example in [17]. At this point it is just to show some naturally occurring lax functors and pseudofunctors.

Example 1.1. A group morphism $\pi : H \rightarrow G$ is a functor. According to Bénabou [3] (see [23]), any functor over a category G corresponds to a normal lax functor $\mathbf{H} : G \rightarrow \text{Mod}^{\text{op}}$. The functor π is a fibration if and only if it is an opfibration if and only if it is Giraud-Conduché if and only if it is a surjective group morphism. In this case, the normal lax functor is actually a pseudofunctor. Let us now describe it. The fibre of the functor π over o_G is the kernel K of the group morphism π ; so $\mathbf{H}o_G = K$ as a one-object category. For $a : o_G \rightarrow o_G$ in G , the module $\mathbf{H}a : K \rightarrow K$ is defined by

$$(\mathbf{H}a)(o_K, o_K) = \{x \in H : \pi(x) = a\} .$$

In particular, $\mathbf{H}1 = K$ is the identity module. For a composable pair $o_G \xrightarrow{b} o_G \xrightarrow{a} o_G$ in G , the composite module $\mathbf{H}b \circ \mathbf{H}a$ is seen to be the set

$$(\mathbf{H}b \circ \mathbf{H}a)(o_K, o_K) = (\{x \in H : \pi(x) = a\} \times \{y \in H : \pi(y) = b\})/K$$

of orbits under the action of K on the right in the first factor of the product and on the left in the second. The composition constraint is the bijection

$$(\mathbf{H}b \circ \mathbf{H}a)(o_K, o_K) \longrightarrow \mathbf{H}(ab)(o_K, o_K) , \quad [x, y] \mapsto xy .$$

A cleavage for the fibration $\pi : H \rightarrow G$ amounts to a splitting σ of π as a function on elements; that is, $\pi\sigma a = a$ for all $a \in G$. The chosen cartesian morphism over $a : o_G \rightarrow o_G$ is $\sigma(a) : o_H \rightarrow o_H$. The pseudofunctor $\mathbf{H} : G \rightarrow \text{Cat}^{\text{op}}$ corresponding to this cloven fibration is defined by $\mathbf{H}o_G = K$, $(\mathbf{H}a)k = k^{\sigma(a)}$ for $a \in G$ and $k \in K$, while the component at o_K of the invertible composition constraint $(Kb)(Ka) \Rightarrow K(ab)$ is equal to $\sigma(a)\sigma(b)\sigma(ab)^{-1} \in K$. We use the same symbol for the pseudofunctors $\mathbf{H}$ since the first is equivalent to the composite of the second with the canonical pseudofunctor $(-)_*$.

The groupoid of automorphisms in G will be denoted by G^{aut} . The objects are elements $a \in G$. A morphism $f : a \rightarrow b$ is an element $f \in G$ such that $fa = bf$. If G_{conj} denotes the G -set whose elements are those of G and whose G -action is by conjugation then there is an equivalence of categories

$$[G, \text{Set}]/G_{\text{conj}} \simeq [G^{\text{aut}}, \text{Set}] . \quad (1.3)$$

Since G_{conj} is a monoid in $[G, \text{Set}]$, there is a monoidal structure on the left hand side of (1.3) whose tensor product takes cartesian product of the morphisms over G_{conj} followed by the monoid multiplication. This monoidal structure is closed (on both sides) and so transports to a promonoidal structure on G^{aut} . Recall from the end of Section 4 of [10] that this promonoidal structure is defined by

$$P(a, b; c) = \{(u, v) \in G \times G : {}^u a {}^v b = c\} \quad \text{and} \quad Jc = \begin{cases} 1 & \text{if } c = 1 \\ \emptyset & \text{if } c \neq 1 \end{cases}$$

and that there is a braiding

$$\gamma_{a,b;c} : P(a, b; c) \xrightarrow{\cong} P(b, a; c) , \quad (u, v) \mapsto ({}^u a v, u) .$$

It also has a twist

$$\tau_a = a : a \rightarrow a .$$

The reader is invited to check the commutativity of (1.4) which is the main twist condition.

$$\begin{array}{ccc} P(a, b; c) & \xrightarrow{\gamma_{a,b;c}} & P(b, a; c) \\ P(1, 1; \tau_c) \downarrow & & \downarrow P(\tau_a, \tau_b; 1) \\ P(a, b; c) & \xleftarrow{\gamma_{b,a;c}} & P(b, a; c) \end{array} \quad (1.4)$$

Furthermore, G^{aut} is a $*$ -autonomous promonoidal category in the sense of [9]: we have the natural isomorphisms

$$\begin{aligned} P(a, b; c^{-1}) &\xrightarrow{\cong} P(b, c; a^{-1}) , & (u, v) &\rightsquigarrow (u^{-1}v, u^{-1}) ; \\ P(a, b; c^{-1}) &\xrightarrow{\cong} P(b^{-1}, a^{-1}; c)^{\text{op}} , & (u, v) &\rightsquigarrow (v, u) . \end{aligned}$$

Proposition 1.2. [24, 10, 11] *The braided monoidale G^{aut} is the monoidal centre of the monoidale G in Mod^{op} . The convolution braided monoidal category $[G^{\text{aut}}, \text{Set}]$ is braided monoidal equivalent to the monoidal centre $\mathfrak{Z}[G, \text{Set}]^1$ of the cartesian monoidal category $[G, \text{Set}]$ of G -sets.*

We note that the equivalence becomes balanced on transport of the convolution twist to the monoidal centre.

Example 1.3. We return to our surjective group morphism $\pi : H \rightarrow G$. Since fibrations in Cat are preserved by 2-functors of the form $[\mathcal{D}, -]$, we have the fibration $\pi^{\text{aut}} : H^{\text{aut}} \rightarrow G^{\text{aut}}$. The corresponding pseudofunctor $\mathbf{H}^{\text{aut}} : G^{\text{aut}} \rightarrow \text{Mod}^{\text{op}}$ is defined as follows. For each $a \in G$, the category $\mathbf{H}^{\text{aut}}a$ has objects those $x \in H$ with $\pi(x) = a$; morphisms $k : x \rightarrow x_1$ are those $k \in K$ with $kx = x_1k$; composition is that of K . For $f : b \rightarrow a$ in G^{aut} , the module $\mathbf{H}^{\text{aut}}f : \mathbf{H}^{\text{aut}}a \rightarrow \mathbf{H}^{\text{aut}}b$ is defined by

$$(\mathbf{H}^{\text{aut}}f)(y, x) = \{h \in H^{\text{aut}}(y, x) : \pi(h) = f\} .$$

The splitting σ for π also gives a cleavage for π^{aut} . The corresponding pseudofunctor $\mathbf{H}^{\text{aut}} : G^{\text{aut}} \rightarrow \text{Cat}^{\text{op}}$ has the same value on objects as in the last paragraph. For $f : b \rightarrow a$ in G^{aut} , the functor $\mathbf{H}^{\text{aut}}f : \mathbf{H}^{\text{aut}}a \rightarrow \mathbf{H}^{\text{aut}}b$ takes $k : x \rightarrow x_1$ in $\mathbf{H}^{\text{aut}}a$ to $k^{\sigma(f)} : x^{\sigma(f)} \rightarrow x_1^{\sigma(f)}$. The invertible composition constraint $(\mathbf{H}^{\text{aut}}g)(\mathbf{H}^{\text{aut}}f) \Rightarrow \mathbf{H}^{\text{aut}}(fg)$ has component at $x \in \mathbf{H}^{\text{aut}}a$ equal to $\sigma(f)\sigma(g)\sigma(fg)^{-1}$.

2. MONOIDALES IN CONVOLUTION BICATEGORIES

One virtue of the promonoidal groupoid G^{aut} over the monoid G_{conj} is that we can obtain convolution balanced monoidal structures on functors from G^{aut} , not only into Set but, into any nice enough monoidal category; or even on pseudofunctors from G^{aut} into any nice enough monoidal bicategory.

Let $\mathcal{K}$ be a monoidal bicategory with coproducts preserved by horizontal composition in each variable. The tensor product will be denoted by $- \otimes - : \mathcal{K} \times \mathcal{K} \rightarrow \mathcal{K}$ with unit object $\mathcal{I}$. Think of G^{aut} as a bicategory with only identity 2-cells. We will make explicit the convolution monoidal structure on the bicategory $\text{Ps}(G^{\text{aut}}, \mathcal{K})$.

¹We are using $\mathfrak{Z}$ for the centre of a monoidal category since we like the fact that it is the first letter of both the German words for centre and for braid.

Take $\mathbb{S}, \mathbb{T} \in \text{Ps}(G^{\text{aut}}, \mathcal{K})$. Put²

$$(\mathbb{S} \star \mathbb{T})c = \sum_{ab=c} \mathbb{S}a \otimes \mathbb{T}b \left(\simeq \int_{\text{ps}}^{a,b} P(a, b; c) \cdot \mathbb{S}a \otimes \mathbb{T}b \right) \quad (2.5)$$

and, for $f : c \rightarrow c_1$ in G^{aut} , define $(\mathbb{S} \star \mathbb{T})f$ by commutativity in

$$\begin{array}{ccc} \mathbb{S}a \otimes \mathbb{T}b & \xrightarrow{\text{in}_{a,b}} & \sum_{ab=c} \mathbb{S}a \otimes \mathbb{T}b \\ \mathbb{S}f \otimes \mathbb{T}f \downarrow & & \downarrow (\mathbb{S} \star \mathbb{T})f \\ \mathbb{S}^f a \otimes \mathbb{T}^f b & \xrightarrow{\text{in}_{f_a, f_b}} & \sum_{a_1 b_1 = c_1} \mathbb{S}a_1 \otimes \mathbb{T}b_1 \end{array} \quad (2.6)$$

This defines the tensor product $\mathbb{S} \star \mathbb{T}$ for a monoidal structure on $\text{Ps}(G^{\text{aut}}, \mathcal{K})$ with unit $\mathbb{J} : G^{\text{aut}} \rightarrow \mathcal{K}$ defined by

$$\mathbb{J}c = \begin{cases} \mathcal{J} & \text{if } c = 1 \\ 0 & \text{if } c \neq 1 \end{cases} \quad (2.7)$$

which becomes functorial on noting that, for $f : c \rightarrow c'$, if $c = 1$ then $c' = 1$.

We can now contemplate monoidales $\mathbb{M}$ in $\text{Ps}(G^{\text{aut}}, \mathcal{K})$. Such a monoidale consists of a pseudofunctor $\mathbb{M} : G^{\text{aut}} \rightarrow \mathcal{K}$ equipped with morphisms

$$I : \mathcal{J} \rightarrow \mathbb{M}1 \quad \text{and} \quad \square_{a,b} : \mathbb{M}a \otimes \mathbb{M}b \rightarrow \mathbb{M}(ab)$$

in $\mathcal{K}$ and invertible 2-cells

$$\begin{array}{ccc} \mathbb{M}a \otimes \mathbb{M}b \otimes \mathbb{M}c & \xrightarrow{1 \otimes \square_{b,c}} & \mathbb{M}a \otimes \mathbb{M}(bc) \\ \square_{a,b} \otimes 1 \downarrow & \xRightarrow[\cong]{\alpha_{a,b,c}} & \downarrow \square_{a,bc} \\ \mathbb{M}(ab) \otimes \mathbb{M}c & \xrightarrow{\square_{ab,c}} & \mathbb{M}(abc) \end{array} \quad (2.8)$$

$$\begin{array}{ccccc} & & \mathbb{M}a & & \\ & \swarrow \mathcal{J} \otimes 1 & \downarrow 1 & \searrow 1 \otimes \mathcal{J} & \\ \mathbb{M}1 \otimes \mathbb{M}a & \xRightarrow[\cong]{\lambda_a} & & \xRightarrow[\cong]{\rho_a} & \mathbb{M}a \otimes \mathbb{M}1 \\ & \searrow \square_{1,a} & \downarrow \square_{a,1} & & \\ & & \mathbb{M}a & & \end{array} \quad (2.9)$$

all subject to pseudonaturality

$$\begin{array}{ccc} \mathbb{M}a \otimes \mathbb{M}b & \xrightarrow{\square_{a,b}} & \mathbb{M}c \\ \mathbb{M}f \otimes \mathbb{M}f \downarrow & \xRightarrow[\cong]{} & \downarrow \mathbb{M}f \\ \mathbb{M}^f a \otimes \mathbb{M}^f b & \xrightarrow{\square_{f_a, f_b}} & \mathbb{M}c' \end{array}, \quad (2.10)$$

modificationality and coherence conditions.

²The coend here is in the pseudo-sense appropriate to bicategories.

Example 2.1. The $\mathbf{H}$ of Example 1.1 is a monoidale in $\text{Ps}(G, \text{Mod}^{\text{op}})$. The monoidal structure is provided by the modules

$$\square : \mathbf{Ho}_G \rightarrow \mathbf{Ho}_G \times \mathbf{Ho}_G \quad \text{and} \quad I : \mathbf{Ho}_G \rightarrow \mathbf{1}$$

defined by

$$\square(o_K, o_K; o_K) = \{(u, v) : u, v \in K\} \quad , \quad \square(k, \ell; m)(u, v) = (muk, mv\ell) \quad , \quad I = !_*$$

3. THE TURAEV-VIRELIZIER STRUCTURES

Definition. [27] A G -graded category over k is a k -linear monoidal category $\mathcal{C}$, with finite direct sums, endowed with a system of pairwise disjoint full k -linear subcategories $\mathcal{C}_a, a \in G$, with finite direct sums, such that

- (a) each object $X \in \mathcal{C}$ splits as a direct sum $\bigoplus_a X_a$ where $X_a \in \mathcal{C}_a$ and a runs over a finite subset of G ;
- (b) if $X \in \mathcal{C}_a$ and $Y \in \mathcal{C}_b$ then $X \otimes Y \in \mathcal{C}_{ab}$;
- (c) if $X \in \mathcal{C}_a$ and $Y \in \mathcal{C}_b$ with $a \neq b$ then $\mathcal{C}(X, Y) = 0$;
- (d) the tensor unit I of $\mathcal{C}$ is in $\mathcal{C}_1$.

Turaev-Virelizier call an object X of a G -graded category $\mathcal{C}$ *homogeneous* when there exists a (necessarily unique) $a \in G$ such that $X \in \mathcal{C}_a$; this a is denoted by $|X|$.

They write $\bar{G}$ for the discrete monoidal category of elements of G with the multiplication as tensor product. They write $\text{Aut}(\mathcal{C})$ for the monoidal category of monoidal endo-equivalences of the monoidal k -linear category $\mathcal{C}$ and monoidal natural isomorphisms; the tensor product is composition of functors.

Definition. [27] A G -crossed category $\mathcal{C}$ is a G -graded category over k equipped with a strong monoidal functor $\varphi : \bar{G} \rightarrow \text{Aut}(\mathcal{C})$ such that $\varphi_a(\mathcal{C}_b) \subseteq \mathcal{C}_{a^{-1}ba}$ for all $a, b \in G$.

Proposition 3.1. *Let $\mathcal{V}$ be the monoidal category of modules over a fixed commutative ring. Then monoidales in $\text{Ps}(G^{\text{aut}}, \mathcal{V}\text{-Cat})$ are equivalent to the G -crossed categories of [27].*

Proof. Take a monoidale $\mathbb{M}$ in $\text{Ps}(G^{\text{aut}}, \mathcal{V}\text{-Cat})$. Let $\mathcal{C}$ be the $\mathcal{V}$ -category obtained by taking the completion, with respect to finite direct sums, of the coproduct $\mathcal{C}_{\text{hom}} = \sum_a \mathbb{M}a$ in $\mathcal{V}\text{-Cat}$. Then $\mathcal{C}$ is a G -graded category with $\varphi_f = \mathbb{M}f^{-1}$.

Conversely, take a G -graded category $\mathcal{C}$ and define $\mathbb{M}a$ to be the full sub- $\mathcal{V}$ -category $\mathcal{C}_a$ of $\mathcal{C}$ consisting of the objects homogeneous over $a \in G$. Then $\mathbb{M}$ is a monoidale in $\text{Ps}(G^{\text{aut}}, \mathcal{V}\text{-Cat})$. $\square$

Corollary 3.2. *With $\mathcal{V}$ as in Proposition 3.1, any monoidale in $\text{Ps}(G^{\text{aut}}, \text{Mod}^{\text{op}})$ delivers a G -crossed category on application of the pseudofunctor (1.1).*

Definition. [27] A G -braided category $\mathcal{C}$ is a G -crossed category $(\mathcal{C}, \varphi)$ equipped with a natural family of isomorphisms

$$\gamma_{X,Y} : X \otimes Y \longrightarrow Y \otimes \varphi_{|Y|}(X) \quad ,$$

for $X, Y \in \mathcal{C}$ and Y homogeneous, subject to **three** axioms.

In the next section, we will see how this fits into our theory of braided monoidales.

4. INTERNAL HOMS, BIDUALS AND BRAIDINGS

If $\mathcal{K}$ is left closed, it is straightforward to see that so too is $\text{Ps}(G^{\text{aut}}, \mathcal{K})$:

$$[\mathbb{T}, \mathbb{U}]a = \prod_b [\mathbb{T}b, \mathbb{U}(ab)] . \quad (4.11)$$

Proposition 4.1. *Suppose in $\mathcal{K}$ that direct sums indexed by the elements of G exist and that each $\mathbb{T}b$ has a left bidual $(\mathbb{T}b)^\vee$. Then $\mathbb{T}$ has a left bidual*

$$\mathbb{T}^\vee a = [\mathbb{T}, \mathbb{J}]a = (\mathbb{T}a^{-1})^\vee \quad (4.12)$$

in $\text{Ps}(G^{\text{aut}}, \mathcal{K})$.

Proof. Taking (4.12) as the definition of $\mathbb{T}^\vee$, we need to prove that the canonical morphism $\mathbb{S} \star \mathbb{T}^\vee \longrightarrow [\mathbb{T}, \mathbb{S}]$ is an equivalence for all $\mathbb{S}$. The component of this canonical pseudonatural transformation at c is

$$\sum_{ab=c} \mathbb{S}a \otimes \mathbb{T}(b^{-1})^\vee \simeq \sum_d \mathbb{S}(cd) \otimes \mathbb{T}(d)^\vee \simeq \sum_d [\mathbb{T}d, \mathbb{S}(cd)] \xrightarrow{\text{canon.}} \prod_d [\mathbb{T}d, \mathbb{S}(cd)]$$

in which the arrow is an equivalence because of our assumption about direct sums. $\square$

Corollary 4.2. *All biduals exist in $\text{Ps}(G^{\text{aut}}, \text{Mod}^{\text{op}})$; that is, the monoidal bicategory is autonomous (also called “compact” or “rigid”).*

If $\mathcal{K}$ is equipped with a braiding $\gamma_{X,Y} : X \otimes Y \rightarrow Y \otimes X$ then we obtain a braiding $\gamma_{\mathbb{S}, \mathbb{T}} : \mathbb{S} \star \mathbb{T} \rightarrow \mathbb{T} \star \mathbb{S}$ on $\text{Ps}(G^{\text{aut}}, \mathcal{K})$ as defined by the commutative pentagon (4.13).

$$\begin{array}{ccccc} \mathbb{S}a \otimes \mathbb{T}b & \xrightarrow{\gamma_{\mathbb{S}a, \mathbb{T}b}} & \mathbb{T}b \otimes \mathbb{S}a & \xrightarrow{\mathbb{T}a \otimes 1} & \mathbb{T}^a b \otimes \mathbb{S}a \\ \text{in}_{a,b} \downarrow & & & & \downarrow \text{in}_{a_b, a} \\ \sum_{ab=c} \mathbb{S}a \otimes \mathbb{T}b & \xrightarrow{\gamma_{\mathbb{S}, \mathbb{T}c}} & \sum_{b'a'=c} \mathbb{T}b' \otimes \mathbb{S}a' & & \end{array} \quad (4.13)$$

If $\mathcal{K}$ is balanced then so too is $\text{Ps}(G^{\text{aut}}, \mathcal{K})$ with twist

$$\theta_{\mathbb{S}, a} = (\mathbb{S}a \xrightarrow{\mathbb{S}a} \mathbb{S}a \xrightarrow{\theta_{\mathbb{S}a}} \mathbb{S}a) .$$

Recall that, if $\mathcal{K}$ is symmetric, we choose its twist to be the identity.

We already have the example G^{aut} of a $\ast$ -autonomous balanced monoidale in Mod^{op} .

Proposition 4.3. *The monoidal bicategory $\text{Ps}(G^{\text{aut}}, \text{Mod}^{\text{op}})$ is tortile.*

Moreover, with $\text{Ps}(G^{\text{aut}}, \mathcal{K})$ a braided monoidal bicategory, according to [24], we can contemplate monoidal centres $\mathfrak{Z}\mathbb{M}$ for monoidales $\mathbb{M}$ therein. Since the centre is a limit, it is formed pointwise in $\mathcal{K}$. From [24], we know that $\mathfrak{Z}\mathbb{M}$ is a braided monoidale in $\text{Ps}(G^{\text{aut}}, \mathcal{K})$.

Example 4.4. The $\mathbf{H}^{\text{aut}}$ of Example 1.3 is a balanced monoidale in $\text{Ps}(G^{\text{aut}}, \text{Mod}^{\text{op}})$. The monoidal structure is provided by the modules

$$\square_{a,b} : \mathbf{H}^{\text{aut}}(ab) \rightarrow \mathbf{H}^{\text{aut}}a \times \mathbf{H}^{\text{aut}}b \quad \text{and} \quad I : \mathbf{H}^{\text{aut}}1 \rightarrow 1$$

defined by

$$\square_{a,b}(x, y; z) = \{(u, v) : u, v \in K, {}^u x {}^v y = z\} \quad \text{and} \quad Iz = \begin{cases} 1 & \text{if } z = 1 \\ \emptyset & \text{if } z \neq 1 \end{cases}.$$

The braiding is

$$\gamma_{x,y;z} : \square_{a,b}(x, y; z) \xrightarrow{\cong} \square_{b,a}(y, x; z), \quad (u, v) \mapsto ({}^u x {}^v u, u).$$

The twist $\tau : \mathbf{H}^{\text{aut}} \rightarrow \mathbf{H}^{\text{aut}}$ is given by $\tau_a = \mathbf{H}^{\text{aut}} a : \mathbf{H}^{\text{aut}} a \rightarrow \mathbf{H}^{\text{aut}} a$. We suspect this example is also tortile [19] (also called “ribbon”).

Let us consider the case of $\mathcal{K} = \mathcal{V}\text{-Cat}$ where $\mathcal{V}$ is a complete cocomplete closed symmetric monoidal category. For $\mathbb{S} \in \text{Ps}(G^{\text{aut}}, \mathcal{V}\text{-Cat})$, $f : a \rightarrow b$ in G^{aut} and $A \in \mathbb{S}$, we put $fA = (\mathbb{S}f)A \in \mathbb{S}b$.

Let $\mathbb{M}$ be a monoidale in $\text{Ps}(G^{\text{aut}}, \mathcal{V}\text{-Cat})$. The tensor product consists of $\mathcal{V}$ -functors $\square_{a,b} : \mathbb{M}a \otimes \mathbb{M}b \rightarrow \mathbb{M}(ab)$. The unit is an object I of $\mathbb{M}1$. The associativity constraint consists of a $\mathcal{V}$ -natural family

$$\alpha_{a,b,c} : (A \square_{a,b} B) \square_{ab,c} C \longrightarrow A \square_{a,bc} (B \square_{b,c} C). \quad (4.14)$$

A braiding for $\mathbb{M}$ consists of a $\mathcal{V}$ -natural family

$$\gamma_{a,b} : A \square_{a,b} B \longrightarrow aB \square_{ab,a} A. \quad (4.15)$$

Proposition 4.5. *In the setting of Proposition 3.1, the braided monoidales in $\text{Ps}(G^{\text{aut}}, \mathcal{V}\text{-Cat})$ are equivalent to the G -braided categories of [27].*

According to Section 3 of [24], since pseudolimits limits are formed pointwise, the monoidal centre $\mathfrak{Z}\mathbb{M}$ of a monoidale $\mathbb{M}$ in $\text{Ps}(G^{\text{aut}}, \mathcal{V}\text{-Cat})$ is constructed as follows. The $\mathcal{V}$ -category $(\mathfrak{Z}\mathbb{M})a$ has objects pairs (A, v) where A is an object of $\mathbb{M}a$ and v is a *half G -braiding* for A consisting of a $\mathcal{V}$ -natural family of isomorphisms

$$v_b : A \square_{a,b} B \longrightarrow aB \square_{ab,a} A \quad (4.16)$$

such that $v_1 : A \square_{a,1} I \longrightarrow I \square_{1,a} A$ transports the right unit constraint into the left unit constraint and the following hexagon commutes.

$$\begin{array}{ccc} & A \square_{a,bc} (B \square_{b,c} C) & \xrightarrow{v} (aB \square_{ab,ac} aC) \square_{a(bc),a} A \\ & \nearrow \alpha & \downarrow \alpha \\ (A \square_{a,b} B) \square_{ab,c} C & & aB \square_{ab,ac} (aC \square_{ac,a} A) \\ & \searrow v \otimes 1 & \uparrow 1 \otimes v \\ & (aB \square_{ab,a} A) \square_{ab,c} C & \xrightarrow{\alpha} aB \square_{ab,ac} (A \square_{a,c} C) \end{array}$$

For $f : a \rightarrow {}^f a$, we have $\mathfrak{Z}\mathbb{M}(A, v) = (fA, v')$ where v'_b for $B \in \mathbb{M}b$ is the composite

$$fA \square_{f_a,b} B \cong f(A \square_{a,{}^{f-1}b} f^{-1}B) \xrightarrow{f v_{f^{-1}B}} f(a f^{-1}B \square_{a(f^{-1}b),a} A) \cong {}^f a B \square_{f_a b, f_a} fA.$$

5. FULL CENTRES

Davydov [6] defined the *full centre* of a monoid M in a (not necessarily braided) monoidal category $\mathcal{E}$ to be a commutative monoid $\mathfrak{z}M$ in the monoidal centre $\mathfrak{Z}\mathcal{E}$ of $\mathcal{E}$ satisfying an appropriate universal property. Street [26] pointed out that the pair $(\mathfrak{Z}\mathcal{E}, \mathfrak{z}M)$ is the monoidal centre of the monoidal $(\mathcal{E}, M)$ in the monoidal bicategory of pointed categories.

We can lift this concept of full centre from the monoidal category level to the monoidal bicategory level. The full centre of a monoidale $\mathbb{M}$ in a monoidal bicategory $\mathcal{K}$ is a braided monoidale $\mathfrak{z}\mathbb{M}$ in the monoidal centre $\mathfrak{Z}\mathcal{K}$ of $\mathcal{K}$ satisfying an appropriate universal property.

Proposition 5.1. (i) *The braided monoidal bicategory $\mathrm{Ps}(G^{\mathrm{aut}}, \mathrm{Mod}^{\mathrm{op}})$ is the monoidal centre of the monoidal bicategory $\mathrm{Ps}(G, \mathrm{Mod}^{\mathrm{op}})$.*
(ii) *The full centre of the monoidale $\mathbf{H}$ in $\mathrm{Ps}(G, \mathrm{Mod}^{\mathrm{op}})$ (see Example 2.1) is the braided monoidale $\mathbf{H}^{\mathrm{aut}}$ (see Example 4.4) in $\mathrm{Ps}(G^{\mathrm{aut}}, \mathrm{Mod}^{\mathrm{op}})$.*

REFERENCES

- [1] BARR, M. AND WELLS, C. *Toposes, Triples and Theories*. Grundlehren der math. Wissenschaften 278, Springer-Verlag, 1985.
- [2] BÉNABOU, J. Introduction to bicategories. *Lecture Notes in Mathematics* 47 (Springer-Verlag, 1967) 1–77. 2
- [3] BÉNABOU, J. *Lectures at Oberwolfach (Germany) and other places* 1972 – 1995. 3
- [4] BRUGUIÉRES, A., LACK, S. AND VIRELIZIER, A. Hopf monads on monoidal categories. *Advances in Mathematics* 227(2) (2011) 745–800.
- [5] CHIKHLADZE, D., LACK, S. AND STREET, R. Hopf monoidal comonads. *Theory Appl. Categ.* 24 No. 19 (2010) 554–563.
- [6] DAVYDOV, A. Centre of an algebra. *Advances in Mathematics* 225 (2010) 319–348 1, 9
- [7] DAY, B.J. On closed categories of functors. in “*Reports of the Midwest Category Seminar IV*”, *Lecture Notes in Mathematics* 137 (Springer, Berlin, 1970) 1–38. 2
- [8] DAY, B.J. AND STREET, R. Monoidal bicategories and Hopf algebroids. *Advances in Math.* 129 (1997) 99–157. 1, 2
- [9] DAY, B.J. AND STREET, R. Quantum categories, star autonomy, and quantum groupoids. in “*Galois Theory, Hopf Algebras, and Semiabelian Categories*”, *Fields Institute Communications* 43 (American Math. Soc. 2004) 187–226. 4
- [10] DAY, B.J., PANCHADCHARAM, E. AND STREET, R. Lax braidings and the lax centre. in “*Hopf Algebras and Generalizations*”, *Contemporary Mathematics* 441 (2007) 1–17. 3, 4
- [11] DAY, B.J. AND STREET, R. Centres of monoidal categories of functors. in “*Categories in Algebra, Geometry and Mathematical Physics*”, *Contemporary Mathematics* 431 (2007) 187–202. 4
- [12] EILENBERG, S. AND KELLY, G.M. Closed categories. *Proceedings of the Conference on Categorical Algebra (La Jolla, 1965)* (Springer-Verlag, 1966) 421–562.
- [13] JOYAL, A. AND STREET, R. An introduction to Tannaka duality and quantum groups. In *Category Theory: Proceedings of the International Conference held in Como, Italy, July 22–28, 1990*, A. Carboni, M. C. Pedicchio, and G. Rosolini, Eds., vol. 1488 of *Lecture Notes in Mathematics*. Springer Berlin Heidelberg, 1991, pp. 413–492.
- [14] JOYAL, A. AND STREET, R. Braided tensor categories. *Adv. Math.* 102 (1993) 20–78.
- [15] KELLY, M. AND STREET, R. Review of the elements of 2-categories. *Lecture Notes in Math.* 420 (1974) 75–103. 2
- [16] KELLY, M., LABELLA, A., SCHMITT, V., AND STREET, R. Categories enriched on two sides. *Journal of Pure and Applied Algebra* 168(1) (2002), 53–98.

- [17] MAIER, J., T. NIKOLAUS, T. AND SCHWEIGERT, C. Equivariant Modular Categories via Dijkgraaf-Witten Theory. arXiv:1103.2963. [3](#)
- [18] MÜGER, M. From subfactors to categories and topology. II. The quantum double of tensor categories and subfactors. *J. Pure Appl. Algebra* 180 (2003) 159–219. [1](#)
- [19] SHUM, M. C. Tortile tensor categories. *Journal of Pure and Applied Algebra* 93 (1994) 57–110. [8](#)
- [20] STREET, R. The formal theory of monads. *Journal of Pure and Applied Algebra* 2(2) (1972) 149–168.
- [21] STREET, R. Two constructions on lax functors. *Cahiers de topologie et géométrie différentielle* 13 (1972) 217–264.
- [22] STREET, R. Absolute colimits in enriched categories. *Cahiers de topologie et géométrie différentielle* 24 (1983) 377–379.
- [23] STREET, R. Powerful functors. <http://web.science.mq.edu.au/~street/Pow.fun.pdf> [3](#)
- [24] STREET, R. The monoidal centre as a limit. *Theory and Applications of Categories* 13(13) (2004) 184–190. [1](#), [4](#), [7](#), [8](#)
- [25] STREET, R. *Quantum Groups: A Path to Current Algebra*. Australian Mathematical Society Lecture Series 19. Cambridge University Press, 2007.
- [26] STREET, R. Monoidal categories in, and linking, geometry and algebra. *Bulletin of the Belgian Mathematical Society Simon Stevin* 19(5) (2012) 769–821. [1](#), [9](#)
- [27] TURAEV, V. AND VIRELIZIER, A. On the graded center of graded categories. *Journal of Pure and Applied Algebra* 217 (2013) 1895–1941. [1](#), [3](#), [6](#), [8](#)
- [28] WEBER, M. Algebraic Kan extensions along morphisms of internal algebra classifiers. *Tbilisi Math. J.* 9(1) (2016) 65–142.

DEPARTMENT OF MATHEMATICS, MACQUARIE UNIVERSITY, NSW 2109, AUSTRALIA
 E-mail address: branik.mg@gmail.com

DEPARTMENT OF MATHEMATICS, MACQUARIE UNIVERSITY, NSW 2109, AUSTRALIA
 E-mail address: ross.street@mq.edu.au